[ACM3-1] [GET USED TO GITHUB/JIRA] Read docs, basic jira/git Created: 20/Aug/26 Updated: 22/Aug/26 Due: 21/Aug/26 Resolved: 22/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Task	**Priority:**	Low
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	20/Aug/26
Rank:	0|i000ah:

[ACM3-2] [EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION Created: 20/Aug/26 Updated: 07/Sep/26 Due: 26/Aug/26 Resolved: 27/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None

Affects versions:	None
Fix versions:	None

Type:	Epic	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Rank:	0|i000ap:
Start date:	20/Aug/26

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-3] [TEAM LEADER] kick-off meeting: schedule meetings, work procedures, roles assignment Created: 20/Aug/26 Updated: 22/Aug/26 Due: 21/Aug/26 Resolved: 22/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000ax:

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-4] [TEAM LEADER] Create and setup Jira, define work types

Created: 20/Aug/26 Updated: 22/Aug/26 Due: 21/Aug/26 Resolved: 22/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0

Labels:	None
Remaining Estimate:	Not Specified
Time Spent:	Not Specified
Original estimate:	Not Specified

Start date:	20/Aug/26
Rank:	0|i000b5:

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-5] [TECH LEAD] Create and setup github repository, folder structures

Created: 20/Aug/26 Updated: 22/Aug/26 Due: 21/Aug/26 Resolved: 22/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		

Original estimate:	Not Specified

Start date:	20/Aug/26
Rank:	0|i000bd:

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-6] [DATA] Collecting and check for reliability of dataset Created: 20/Aug/26 Updated: 26/Aug/26 Due: 23/Aug/26 Resolved: 26/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Lê Thanh Lâm
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26

Rank:	0|i000bh:

Description

Scope

Collect and verify the **3 selected healthcare datasets** and their data provenance before conducting exploratory data analysis.

The task includes:

- Identifying and documenting the original source of each dataset.
- Collecting the selected datasets in their raw form.
- Verifying that the downloaded data matches the documented dataset.
- Verifying the target variable and prediction task.
- Verifying the dataset schema and feature definitions against the source documentation.
- Checking file integrity and readability.
- Recording dataset versions or release information when available.
- Reviewing dataset licensing or usage conditions when applicable.
- Recording known limitations or collection-related issues reported by the source.
- Assessing whether each dataset is suitable for the defined research scope.

Definition of Done

- All selected datasets are obtained from documented sources.
- Dataset source and provenance are recorded.
- Dataset version or release information is recorded when available.
- Raw datasets are preserved without modification.
- Dataset files can be loaded successfully.
- Target variable is identified and verified against the source documentation.
- Feature definitions are documented based on the original source.
- Dataset schema is verified against the source documentation.
- Dataset licensing or usage requirements are reviewed when applicable.
- Known dataset limitations reported by the source are documented.
- Dataset suitability for the research scope is confirmed.
- Data collection and validation information is documented.

- QA/Reviewer has reviewed the dataset sources and validation results.

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-7] [PIPELINE] Design ML pipeline workflow Created: 20/Aug/26 Updated: 26/Aug/26 Due: 25/Aug/26 Resolved: 26/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Văn Vĩ Nguyễn
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000bl:

Description

Scope

Design the end-to-end machine learning experiment workflow for the healthcare risk prediction research, defining the logical flow and interactions between data preparation, model training, evaluation, cross-validation, and SHAP-based analysis.

The task includes:

- Define the overall ML experiment workflow.
- Define the inputs and outputs of each pipeline stage.
- Define the cross-validation workflow.
- Define the model training and evaluation flow for AdaBoost, XGBoost, and LightGBM.
- Define the SHAP analysis flow across cross-validation folds.
- Create a pipeline diagram representing the finalized workflow.

Definition of Done

- End-to-end ML pipeline workflow is clearly defined.
- Pipeline stages and their dependencies are identified.
- Input and output of each major stage are documented.
- Data loading and preprocessing flow is defined.
- Cross-validation flow is defined.
- AdaBoost, XGBoost, and LightGBM are represented in the workflow.
- Evaluation stage is represented in the workflow.
- SHAP analysis and cross-fold stability analysis are represented.
- Pipeline diagram is created.
- Diagram clearly shows the data flow and experiment flow.
- Pipeline design is reviewed and approved by the Tech Lead.

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-11] [MODEL] Review and choose tree-based models for project Created: 20/Aug/26 Updated: 22/Aug/26 Due: 22/Aug/26

Resolved: 22/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000bn:

[ACM3-9] [EPIC 2] DATA UNDERSTANDING & PREPARATION Created: 20/Aug/26 Updated: 07/Sep/26 Due: 30/Aug/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Epic	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Lê Thanh Lâm
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Rank:	0|i000bp:
Start date:	21/Aug/26

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-18] [MODEL] Choose parameters, hyperparameters for models, write Related work + experimental design in report Created: 22/Aug/26 Updated: 27/Aug/26 Due: 26/Aug/26 Resolved: 27/Aug/26

Status:	Done

Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000br:

Description

Scope

Define the model configurations and experimental settings for AdaBoost, XGBoost, and LightGBM, and document the corresponding experimental design in the research report.

The task includes:

- Identify the key parameters and hyperparameters for each model.
- Define a consistent hyperparameter selection strategy across models.
- Determine the initial/default hyperparameter values used in the experiments.
- Document the reason why for important hyperparameter choices.
- Define the model training configuration and experimental settings.
- Define the evaluation settings used for model comparison.
- Document the experimental design in the report.

- Review relevant literature on tree-based models and their applications to healthcare risk prediction for the Related Work section.

Definition of Done

Model Configuration

- Key hyperparameters for AdaBoost are identified and documented.
- Key hyperparameters for XGBoost are identified and documented.
- Key hyperparameters for LightGBM are identified and documented.
- Hyperparameter values are defined and recorded in `config.yaml`.
- The same experimental protocol is applied consistently across all models.
- The reason for important hyperparameter choices is documented.
- Random seed and other reproducibility-related settings are defined.

Experimental Design

- Model training procedure is documented.
- Evaluation metrics are documented.
- Experimental settings are consistent with the research questions.
- Experimental design is reviewed by the Tech Lead.

Report

- Relevant literature is collected and reviewed for the Related Work section.
- Related Work section is drafted with appropriate citations.
- Experimental Design section is drafted.
- Model configurations and experimental settings are clearly described.
- Report content is consistent with the implemented experiment configuration.
- Report changes are reviewed.

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-19] [TECH LEAD] write introduction, RQ in report, DOD, task description

Created: 23/Aug/26 Updated: 25/Aug/26 Due: 24/Aug/26 Resolved: 25/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	Low
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000bs:

[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION (ACM3-2)

[ACM3-8] [QA] Review project scope and methodology Created: 20/Aug/26 Updated: 27/Aug/26 Due: 25/Aug/26 Resolved: 27/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 1] PROJECT INIT, TEAM ALIGNMENT & RESEARCH DEFINITION

Type:	Task	**Priority:**	Low
Reporter:	Thanh Le quang	**Assignee:**	Tân Dư
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000bt:

Comments

Comment by Tân Dư [26/Aug/26]
https://github.com/AIVIETNAM-AIO-tlee/healthcare-risk-prediction/blob/91925ce90758009aa49bcc076e11933ad84b02b9/docs/qa-scope-methodology-review-handoff.md

[EPIC 2] DATA UNDERSTANDING & PREPARATION (ACM3-9)

[ACM3-12] [DATA] Do exploratory data analysis on healthcare dataset Created: 20/Aug/26 Updated: 07/Sep/26 Due: 23/Aug/26

Resolved: 26/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 2] DATA UNDERSTANDING & PREPARATION

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Lê Thanh Lâm
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000bx:

Description

Scope

Perform exploratory data analysis (EDA) on the selected healthcare datasets to understand their characteristics, distributions, relationships, and potential data quality issues relevant to the research.

The task includes:

- Summarize the structure and characteristics of each dataset.
- Analyze the distribution of the target variable.
- Analyze feature distributions and feature types.
- Identify potential outliers and unusual observations.
- Analyze relationships between features and the target variable.
- Analyze correlations between numerical features.
- Identify potential class imbalance.
- Compare important characteristics across the selected datasets.
- Produce visualizations and statistical summaries to support later preprocessing and modeling decisions.

Definition of Done

- EDA is completed for all selected healthcare datasets.
- Dataset structure and basic statistics are summarized.
- Target variable distribution is analyzed.
- Feature distributions are analyzed.
- Numerical and categorical features are identified.
- Potential outliers are investigated.
- Class imbalance is identified and reported where applicable.
- Feature-target relationships are analyzed.
- Correlations between relevant numerical features are analyzed.
- Observations and findings are documented.
- EDA findings are linked to relevant preprocessing decisions.
- EDA notebooks/code are reproducible.
- Results are reviewed by the QA/Reviewer and Tech lead.

[EPIC 2] DATA UNDERSTANDING & PREPARATION (ACM3-9)

[ACM3-13] [DATA] Implement preprocess on 3 datasets: missing values, outliers, encode, scaling Created: 20/Aug/26 Updated: 07/Sep/26 Due: 25/Aug/26 Resolved: 26/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 2] DATA UNDERSTANDING & PREPARATION

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Lê Thanh Lâm
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000bz:

Description

Scope

Implement the preprocessing pipeline for the **3 selected healthcare datasets** before model training.

The task includes:

- Handle missing values.
- Detect and handle relevant outliers based on EDA findings.

- Encode categorical features where required.
- Apply feature scaling where appropriate.
- Separate features and target variables.
- Ensure preprocessing is fitted only on training data within each cross-validation fold.
- Provide a consistent preprocessing interface for all 3 datasets.
- Document dataset-specific preprocessing decisions.
- Ensure the preprocessing pipeline is reproducible.

Definition of Done

- Preprocessing pipeline is implemented for Dataset 1.
- Preprocessing pipeline is implemented for Dataset 2.
- Preprocessing pipeline is implemented for Dataset 3.
- Missing values are handled.
- Relevant outliers are handled based on EDA findings.
- Categorical features are encoded where required.
- Numerical features are scaled where appropriate.
- Features and target variables are correctly separated.
- Dataset-specific preprocessing differences are documented.
- Preprocessing is fitted only on training data within each CV fold.
- No data leakage is introduced during preprocessing.
- Preprocessing code is documented and tested.
- Pull Request is reviewed and merged.

[EPIC 2] DATA UNDERSTANDING & PREPARATION (ACM3-9)

[ACM3-23] [DATA] Enhance preprocessing: handle class imbalance, outliers, and feature reduction Created: 28/Aug/26 Updated: 07/Sep/26 Due: 30/Aug/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 2] DATA UNDERSTANDING & PREPARATION

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Lê Thanh Lâm
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	27/Aug/26
Rank:	0|i000c0:

Description

Scope

- Review the current preprocessing pipeline for the three healthcare datasets.
- Analyze and handle **class imbalance** appropriately for each dataset.
- Identify and handle **outliers** in numerical features using an appropriate statistical method.
- Investigate irrelevant/redundant features through EDA.
- Apply **PCA or another justified feature-reduction method** where appropriate.
- Ensure that preprocessing remains consistent and reproducible across the three datasets.
- Update the preprocessing implementation in `src/`.

- Document preprocessing decisions and dataset-specific differences.

Definition of Done

- Class distribution is analyzed for all three datasets.
- An appropriate imbalance-handling strategy is implemented and justified.
- Outliers are identified and handled using a documented method.
- Potentially unnecessary/redundant features are investigated through EDA.
- PCA/feature reduction is applied only when justified by the analysis.
- Preprocessing works correctly for all three datasets.
- No data leakage is introduced during preprocessing.
- Updated preprocessing code is implemented in `src/`.
- Results/figures relevant to preprocessing are saved for the report.
- Changes are documented and ready for review from Tech Lead.

[EPIC 2] DATA UNDERSTANDING & PREPARATION (ACM3-9)

[ACM3-15] [QA] Validate preprocessing and cross-validation leakage, write tests Created: 20/Aug/26 Updated: 07/Sep/26

Due: 26/Aug/26 Resolved: 27/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 2] DATA UNDERSTANDING & PREPARATION

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Tân Dư
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	21/Aug/26
Rank:	0|i000c0:i

Description

Scope

Validate the correctness and reliability of the preprocessing, dataset splitting, and Stratified K-Fold cross-validation pipeline.

The task focuses on identifying data leakage, incorrect data transformations, inconsistent preprocessing behavior, and reproducibility issues before the models are evaluated.

The task includes:

- Review the preprocessing pipeline for all **3 healthcare datasets**.
- Verify missing-value handling, encoding, scaling, and outlier handling.
- Verify that preprocessing is fitted only on training data within each fold.
- Validate train/validation/test separation.
- Validate Stratified K-Fold implementation.
- Check that target information is not leaked into input features.
- Check that validation and test data do not influence preprocessing.
- Verify that preprocessing produces consistent outputs.
- Verify reproducibility using the configured random seed.
- Document identified issues, test results, and corrective actions.

Definition of Done

Preprocessing Validation

- Missing-value handling is tested for each applicable dataset.
- Encoding behavior is tested for categorical features.
- Scaling behavior is tested for numerical features.
- Outlier handling is tested where applicable.
- Feature and target separation is verified.
- Preprocessing does not modify the original raw datasets.
- Preprocessing output has the expected feature dimensions.
- Preprocessing produces consistent results for repeated runs with the same configuration.

Data Leakage Validation

- Imputation is fitted only on training data.
- Encoder is fitted only on training data.
- Scaler is fitted only on training data.
- Outlier detection/handling parameters are fitted only on training data.
- Validation data is not used to fit preprocessing components.
- Test data is not used during model training or preprocessing fitting.
- Target information is not included in input features.
- No duplicated samples exist across train and validation/test splits where applicable.
- Cross-validation folds do not contain overlapping samples.
- No information from future or validation/test samples is used during training.

Cross-Validation Validation

- Stratified K-Fold generates the configured number of folds.
- Each sample is used as validation data exactly once across the folds.
- Training and validation indices do not overlap within a fold.
- Class proportions are reasonably preserved across folds.
- Test data remains completely outside the K-Fold process.
- Fold generation is reproducible using the configured random seed.
- The same splitting protocol is applied consistently across datasets.

Test Cases

- Unit tests are implemented for preprocessing components.
- Unit tests are implemented for dataset splitting.
- Unit tests are implemented for Stratified K-Fold generation.
- Tests cover normal/expected inputs.
- Tests cover missing-value cases.
- Tests cover invalid or unexpected input cases where applicable.
- Tests verify train/validation/test separation.
- Tests verify that preprocessing is fitted only on training data.
- Tests verify reproducibility with the same random seed.
- Integration test verifies the complete preprocessing -> split → CV pipeline.
- All automated tests pass successfully.

Comments

Comment by Tân Dư [26/Aug/26]
https://github.com/AIVIETNAM-AIO-tlee/healthcare-risk-prediction/pull/3

[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON (ACM3-10)

[ACM3-24] [QA] Implement tests for data preprocessing, models, model results

Created: 28/Aug/26 Updated: 07/Sep/26 Due: 03/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None

Fix versions:	None
Parent:	[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Tân Dư
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	01/Sep/26
Rank:	0|i000c0:r

Description

Scope

- Test class imbalance handling.
- Test outlier handling.
- Test feature encoding and scaling.
- Test PCA/feature reduction where applicable.
- Test preprocessing consistency across the three datasets.
- Test AdaBoost, XGBoost, and LightGBM training and prediction.
- Test evaluation metrics: ROC-AUC, PR-AUC, Recall, and F1.
- Validate model results and output formats.
- Check potential preprocessing and cross-validation data leakage.

Definition of Done

- Test cases for class imbalance handling are implemented.
- Test cases for outlier handling are implemented.
- Tests verify preprocessing transformations produce valid outputs.

- Tests verify encoding and scaling behave as expected.
- Tests verify PCA/feature reduction produces the expected output when applied.
- Tests cover all three healthcare datasets where applicable.
- Tests verify AdaBoost, XGBoost, and LightGBM can be trained successfully.
- Tests verify prediction outputs have the expected shape and valid values.
- Evaluation metrics return valid results within their expected ranges.
- Tests verify that the evaluation results are generated correctly.
- Tests cover potential data leakage in preprocessing and cross-validation.
- All implemented tests pass successfully.
- Test results are documented.
- The task is ready for review.

Comments

Comment by Tân Dư [04/Sep/26]
https://github.com/AIVIETNAM-AIO-tlee/healthcare-risk-prediction/pull/10 Thanh Le quang

[ACM3-10] [EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON Created: 20/Aug/26 Updated: 07/Sep/26 Due: 03/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Epic	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran

Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Rank:	0|i000c1:
Start date:	22/Aug/26

[EPIC 2] DATA UNDERSTANDING & PREPARATION (ACM3-9)

[ACM3-14] [PIPELINE] Implement dataset spliting and stratified K-fold cross-validation

Created: 20/Aug/26 Updated: 27/Aug/26 Due: 25/Aug/26 Resolved: 27/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 2] DATA UNDERSTANDING & PREPARATION

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Văn Vĩ Nguyễn
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		

Time Spent:	Not Specified
Original estimate:	Not Specified

Start date:	24/Aug/26
Rank:	0|i000c9:

Description

Scope

Implement the dataset splitting and Stratified K-Fold cross-validation strategy used in the healthcare risk prediction experiments.

The task includes:

- Define the train/test splitting strategy.
- Implement Stratified K-Fold cross-validation on the training data.
- Preserve the target class distribution across folds.
- Ensure preprocessing is performed within each training fold.
- Provide reproducible data splitting using a configurable random seed.
- Ensure the same cross-validation protocol is applied consistently across all three healthcare datasets and three models.
- Provide fold indices or split information for downstream model training, evaluation, and SHAP analysis.

Definition of Done

- Train/test splitting strategy is implemented.
- Stratified K-Fold cross-validation is implemented.
- The number of folds is configurable.
- Random seed is configurable and recorded.
- Class distribution is preserved across validation folds.
- Test data is kept separate from the cross-validation process.
- The same split strategy is applied consistently across all datasets.
- Fold indices are reproducible.
- Fold information can be used for per-fold SHAP analysis.

- Code is integrated into the ML pipeline.
- Pull Request is reviewed and merged.

[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON (ACM3-10)

[ACM3-20] [MODEL] Implement Adaboost, XGBoost, LightGBM Created: 24/Aug/26 Updated: 07/Sep/26 Due: 27/Aug/26 Resolved: 27/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	22/Aug/26
Rank:	0|i000cd:

Description

Scope

Implement AdaBoost, XGBoost, and LightGBM as the three tree-based models used in the healthcare risk prediction experiments.

The implementations should:

- Work with all selected healthcare datasets.
- Allow model configurations to be loaded from `config.yaml`.
- Provide reproducible model training using a configurable random seed.
- Return prediction probabilities required for downstream evaluation metrics and SHAP analysis.

Definition of Done

- AdaBoost implementation is completed.
- XGBoost implementation is completed.
- LightGBM implementation is completed.
- All three models follow a consistent interface.
- Model hyperparameters are configurable.
- All models can be integrated into the cross-validation pipeline.
- All models work correctly with the selected healthcare datasets.
- Prediction probabilities are available for ROC-AUC, PR-AUC, and SHAP analysis.
- Code is reviewed and Pull Request is merged.

[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON (ACM3-10)

[ACM3-21] [MODEL] Implement evaluation metrics (ROC-AUC, PR-AUC, Recall, F1) and give comparisons.

Created: 24/Aug/26 Updated: 07/Sep/26 Due: 27/Aug/26 Resolved: 27/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	22/Aug/26
Rank:	0|i000cf:

Description

Scope

Implement a standardized evaluation module for comparing AdaBoost, XGBoost,
and LightGBM across the selected healthcare datasets.

The evaluation should include:

- ROC-AUC
- PR-AUC
- Recall

- F1-score

The module should:

- Calculate metrics consistently across all models and datasets.
- Calculate metrics for each cross-validation fold.
- Aggregate fold-level results using mean and standard deviation.
- Generate a standardized comparison table for model performance.
- Store evaluation results in a reproducible format.

Definition of Done

- ROC-AUC, PR-AUC, Recall, and F1-score are implemented correctly.
- The same evaluation procedure is applied to AdaBoost, XGBoost, and LightGBM.
- Metrics are calculated for all selected healthcare datasets.
- Metrics are calculated for each cross-validation fold.
- Mean ± standard deviation are reported across folds.
- A standardized model comparison table is generated.
- Code and evaluation procedure are documented.
- Pull Request is reviewed and merged.

[ACM3-16] [EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS Created: 20/Aug/26 Updated: 07/Sep/26 Due: 03/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Epic	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Văn Vĩ Nguyễn
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Rank:	0|i000ch:
Start date:	27/Aug/26

[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON (ACM3-10)

[ACM3-22] [MODEL] Create config.yaml for reproducible model experiments
Created: 24/Aug/26 Updated: 07/Sep/26 Due: 27/Aug/26 Resolved: 27/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 3] ML PIPELINE & TREE-BASED MODEL COMPARISON

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran

Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	22/Aug/26
Rank:	0|i000cl:

Description

Scope

- Dataset configuration
- Model configuration
- Cross-validation settings
- Random seed
- Evaluation metrics
- SHAP settings

Definition of Done

- config.yaml is created
- Pipeline can load config.yaml
- Random seed is configurable
- Model settings are configurable
- Configuration is tested
- Documentation is updated
- PR is reviewed and merged

[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS (ACM3-16)

[ACM3-25] [MODEL] Implement SHAP explainability Created: 28/Aug/26 Updated: 07/Sep/26 Due: 01/Sep/26 Resolved: 30/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	27/Aug/26
Rank:	0|i000cn:

Description

Scope

- Integrate SHAP into the existing model pipeline.
- Generate SHAP values from trained models.
- Calculate global feature importance.

- Generate SHAP summary/feature importance visualizations.
- Ensure the implementation is compatible with the existing preprocessing and cross-validation pipeline.
- Save SHAP outputs for use in the stability analysis.

Definition of Done

- SHAP is successfully integrated with AdaBoost, XGBoost, and LightGBM.
- SHAP values are generated correctly for the required trained models.
- SHAP values have the correct dimensions corresponding to samples and features.
- Global SHAP feature importance is calculated.
- Feature importance rankings are generated.
- SHAP summary plots/visualizations are generated.
- The implementation supports the three healthcare datasets.
- SHAP outputs are saved in a reusable format for the stability analysis.
- The implementation does not use the held-out test set for model/explanation selection.
- Code is documented and reproducible.
- Implementation is ready for QA review.

[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS (ACM3-16)

[ACM3-26] [MODEL/PIPELINE] Implement SHAP stability analysis across CV folds Created: 28/Aug/26 Updated: 07/Sep/26

Due: 01/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None

Fix versions:	None
Parent:	[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS

Type:	Task	**Priority:**	High
Reporter:	Thanh Le quang	**Assignee:**	Văn Vĩ Nguyễn
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	27/Aug/26
Rank:	0|i000co:

Description

Scope

- Collect SHAP feature importance rankings from each CV fold.
- Compare feature rankings across folds.
- Implement an appropriate ranking stability measure, such as **Spearman rank correlation**.
- Optionally analyze **Top-K feature overlap**.
- Compare SHAP stability across models and datasets.
- Generate tables and visualizations supporting RQ3.

Definition of Done

- SHAP feature rankings are collected for all relevant CV folds.
- Ranking comparison across the 5 folds is implemented.
- A justified SHAP stability metric is implemented.
- Spearman rank correlation (or other measures) is calculated correctly if selected.
- Top-K feature overlap is calculated if selected.
- Stability results can be compared across AdaBoost, XGBoost, and LightGBM.

- Stability results can be compared across the three healthcare datasets.
- Appropriate tables/visualizations are generated.
- Results are saved in a reproducible format.
- The analysis uses CV-fold results rather than the held-out test set.
- Results are sufficient to support **RQ3**.
- Implementation is ready for QA and Teach lead review.

[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS (ACM3-16)

[ACM3-28] [QA] Test SHAP stability analysis Created: 28/Aug/26 Updated: 07/Sep/26 Due: 03/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Tân Dư
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	01/Sep/26
Rank:	0|i000co:i

Description

Scope

Validate the correctness of SHAP feature importance stability analysis across cross-validation folds.

Definition of Done

- SHAP rankings from all required CV folds are validated.
- Ranking comparison across folds is tested.
- Spearman rank correlation calculation is tested if used.
- Top-K feature overlap calculation is tested if used.
- Stability scores are validated against expected results on controlled/test data.
- Tests verify that the held-out test set is not used for SHAP stability analysis.
- Results are reproducible under the same random seed/configuration.
- All tests pass successfully.
- Test results are documented.

Comments

Comment by Tân Dư [04/Sep/26]
https://github.com/AIVIETNAM-AIO-tlee/healthcare-risk-prediction/pull/10 Thanh Le quang

[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS (ACM3-16)

[ACM3-27] [QA] Test SHAP explainability Created: 28/Aug/26 Updated: 07/Sep/26 Due: 03/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 4] K-FOLD VALIDATION & SHAP ANALYSIS

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Tân Dư
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	01/Sep/26
Rank:	0|i000co:r

Description

Scope

Validate the correctness of SHAP value generation and feature importance analysis for the supported tree-based models.

Definition of Done

- SHAP value generation is tested.
- SHAP output dimensions are validated.
- SHAP values correspond correctly to input features.
- Global feature importance calculation is tested.
- Feature importance rankings are validated.
- SHAP outputs are tested for AdaBoost, XGBoost, and LightGBM.

- SHAP visualizations/output are validated where applicable.
- Tests pass successfully.
- Test results are documented.

Comments

Comment by Tân Dư [04/Sep/26]
https://github.com/AIVIETNAM-AIO-tlee/healthcare-risk-prediction/pull/10 Thanh Le quang

[ACM3-17] [EPIC 5] RESULTS, REPORT & FINAL REVIEW Created: 20/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Epic	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Rank:	0|i000cp:
Start date:	28/Aug/26

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-38] [REPORT-QA] Write test in subsection test in Data & experimental design section Created: 29/Aug/26

Updated: 07/Sep/26 Due: 03/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Tân Dư
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	01/Sep/26
Rank:	0|i000cy:

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-37] [TEAM LEADER] Final review report, code, docs

Created: 28/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	28/Aug/26
Rank:	0|i000d1:2

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-35] [SLIDE] Prepare presentation structure and research overview Created: 28/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	28/Aug/26
Rank:	0|i000d1:4

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-34] [REPORT] Write Research Architecture and General Pipeline Created: 28/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26

Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Thanh Le quang
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	28/Aug/26
Rank:	0|i000d1:9

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-33] [REPORT-PIPELINE] Write SHAP Stability Methodology Created: 28/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26

Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Văn Vĩ Nguyễn
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	28/Aug/26
Rank:	0|i000d1:i

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-32] [REPORT-MODEL] Write SHAP Explainability Methodology

Created: 28/Aug/26 Updated: 30/Aug/26 Due: 05/Sep/26 Resolved: 30/Aug/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None

Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Dang Tran
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	28/Aug/26
Rank:	0|i000d2:

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-31] [REPORT-PIPELINE] Write dataset splitting and cross-validation methodology

Created: 28/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Văn Vĩ Nguyễn
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	28/Aug/26
Rank:	0|i000d3:

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-30] [REPORT-DATA] Write data preprocessing methodology Created: 28/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26

Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Lê Thanh Lâm

Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Start date:	28/Aug/26
Rank:	0|i000d5:

[EPIC 5] RESULTS, REPORT & FINAL REVIEW (ACM3-17)

[ACM3-29] [REPORT-DATA] Write dataset description and EDA Created: 28/Aug/26 Updated: 07/Sep/26 Due: 05/Sep/26 Resolved: 07/Sep/26

Status:	Done
Project:	AIO-CONQUER MODULE 3
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	[EPIC 5] RESULTS, REPORT & FINAL REVIEW

Type:	Task	**Priority:**	Medium
Reporter:	Thanh Le quang	**Assignee:**	Lê Thanh Lâm
Resolution:	Done	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		

Time Spent:	Not Specified
Original estimate:	Not Specified

Start date:	28/Aug/26
Rank:	0|i000d9: